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Research paper

Chaoborus predation and delayed reproduction in *Daphnia*: a demographic modeling approach

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Abstract. I develop a demographic model that examines the impact of *Chaoborus* predation on the population dynamics and life history of *Daphnia*. Predation effects are determined through analysis of the various components of the predator-prey interaction (encounter, attack, strike efficiency), and are integrated into a stage-classified matrix population model. The model is parameterized with data from interactions between *D. pulex* and fourth-instar *C. americanus*. I test this model with two laboratory experiments that examine population growth of *D. pulex* under the influence of five different levels of *Chaoborus* predation. With the exception of a single predation treatment in each experiment, the model accurately predicted the observed reduction in *Daphnia* numbers with increasing *Chaoborus* predation. I then use this model to investigate the evolution of delayed reproduction in *D. pulex* that are exposed to *Chaoborus*. I ask whether delayed reproduction may evolve in *Daphnia* that are subjected to *Chaoborus* predation as a trade-off for the benefits of larger body size. The model predicts that the effectiveness of such a life history trade-off depends on the relative sizes of predator and prey. In some interactions between *Chaoborus* and *Daphnia*, increasing *Daphnia* body length by as little as 5% from base growth trajectories sufficiently increases fitness (by reducing vulnerability to *Chaoborus* predation) to compensate for the cost of delayed reproduction. In other interactions, however, increased body length provides no benefit to *Daphnia* (and may even reduce fitness), and selection would act against the evolution of delayed reproduction.

Key words: body size, *Chaoborus americanus*, *Daphnia pulex*, delayed reproduction, demographic model, induced antipredator defenses, life history, size-selective predation, trade-offs, zooplankton

Introduction

An organism's life history is composed of an integrated suite of demographic traits that include such features as age at maturity, size at maturity, clutch size, and offspring size (Stearns, 1992; Roff, 1992). The evolution of a particular life history strategy, a coordinated set of life history traits that evolve together (Stearns, 1992) is shaped by natural selection through the relative allocation of resources to growth and reproduction in ways that maximize fitness in particular environmental settings. Trade-offs between life history traits (e.g., age at maturity vs. size at maturity, clutch size vs. egg size) play an integral role in this

process, and serve to mold and constrain the form of a life history strategy under a specified set of selection pressures.

Predation, through its effect on age- or stage-specific survivorship, exerts a powerful influence on the life histories of prey (Law, 1979). Various prey life history traits are altered in the presence of predators, either as a direct, facultative shift (adaptation) to reduce predation pressure or as an indirect consequence (cost) of other antipredator responses, such as morphological or behavioral changes (Ball and Baker, 1996). These induced responses are especially evident in the water flea *Daphnia* (Crustacea: Cladocera), a common and important constituent of the zooplankton communities of lakes and ponds, which exhibits a variety of life history changes (as well as morphological and behavioral alterations) in response to a diverse array of fish and invertebrate predators (Tollrian, 1995b; Boersma *et al*., 1998; Riessen, 1999). These life history shifts are mediated by chemical cues (kairomones) released by the predators, which allow *Daphnia* to gauge potential predation risk and react accordingly.

Predation on zooplankton is strongly size-selective, which results in survivorship being reduced more in certain ages or stages than in others. Planktivorous fish are generally visual predators and select larger, more conspicuous prey (Brooks and Dodson, 1965; O'Brien, 1979, 1987; Greene, 1983). Most invertebrate predators, on the other hand, select smaller prey because of the relatively low predator/prey size ratio, which makes handling of larger prey difficult (Dodson, 1974; Greene, 1983; Gliwicz and Umana, 1994), although *Chaoborus* larvae preferentially consume intermediate-sized zooplankton prey (see below). These patterns of mortality result in strong selection pressures that influence growth rates and body sizes of the various juvenile and adult stages, which in turn affect other life history traits, such as age at maturity and clutch size.

In this paper I examine the impact of predation by larvae of the phantom midge *Chaoborus* (Insecta: Diptera) on life history evolution in *D. pulex* Leydig, a moderately large daphnid species that is most commonly found in temporary and permanent ponds, especially those that lack fish. *Daphnia pulex* responds to kairomones released by *Chaoborus* by developing neck spines and an altered vertical migration behavior, which reduce mortality from this size-selective predator (Krueger and Dodson, 1981; Havel and Dodson, 1984; Dodson, 1988, 1989; Tollrian, 1995a; Spitze and Sadler, 1996; Nesbitt *et al*., 1996). The induced development of neck spines in juvenile *D. pulex* as an antipredator defense is typically accompanied by delayed reproduction (increased age at maturity and increased duration of adult instars) (Riessen and Sprules, 1990; Tollrian, 1995b; Riessen, 1999), although this association is not universal for all clones (Walls *et al*., 1991; Spitze, 1992). Delayed reproduction by itself reduces population growth rate, r, and thus results in a decrease in fitness. This life history change has primarily been interpreted as the cost of neck spine formation and thus the cost of reducing vulnerability to *Chaoborus*

predation (Havel and Dodson, 1987; Walls and Ketola, 1989; Black and Dodson, 1990; Riessen and Sprules, 1990; Riessen, 1992).

Several studies, however, have questioned the direct linkage of delayed reproduction to neck spine induction (Lüning, 1992, 1994, 1995; Black, 1993; Tollrian, 1995b; Repka and Pihlajamaa, 1996), and two of these (Black, 1993; Tollrian, 1995b) have proposed an alternative hypothesis for this observed delay in reproduction by *D. pulex*. Both Black (1993) and Tollrian (1995b) observed that *D. pulex* exposed to *Chaoborus* kairomones in their experiments exhibited an increase in body size, as well as the formation of neck spines and delayed reproduction. They proposed that the delayed reproduction is not the cost of producing neck spines, but rather the cost of an increased body size, which itself is an adaptation to reduce *Chaoborus* predation. This integrated life history shift (larger body size at the expense of later reproduction) is thus viewed as a separate response to *Chaoborus* kairomones, independent from that which results in neck spine induction. Increased body size may result from a prolonged period of development at each instar, during which additional resources may be acquired for growth. Increased body size may alternatively result from a shift in the allocation of resources from reproduction to somatic growth, which would consequently result in delayed reproduction.

I develop a demographic model that examines the effect of predation by *Chaoborus americanus* (Johannsen) on the population dynamics and life history of *Daphnia pulex*, and employ it to investigate the evolution of delayed reproduction in *Daphnia* that are exposed to *Chaoborus* predation. This predator–prey interaction, which constitutes a major source of mortality for *D. pulex* in the fishless ponds and lakes in which these species co-occur (Lynch, 1979; Spitze, 1991), is one of the best studied and best understood in aquatic habitats (Swift and Fedorenko, 1975; Riessen *et al.* 1984; Havel and Dodson, 1984; Dodson, 1989). The model specifically predicts *Chaoborus*-induced size-dependent mortality on *D. pulex* populations and its effect on *Daphnia* population growth rates (r). My approach is to analyze individual components (conditional probabilities) of the predator–prey interaction [predation cycle: encounter, attack, capture, and ingestion (O'Brien, 1979, 1987)] to calculate these predation effects. The model thus connects mechanisms at the individual level directly to the dynamics of the *Daphnia* population, allowing a very detailed and quantitative understanding of the effects of predation on this species. I use the value of r generated by the model as a measure of the fitness of a particular suite of *Daphnia* life history traits (Partridge and Harvey, 1988; Stearns, 1992; McGraw and Caswell, 1996).

I first develop the general form of the demographic model for the *Chaoborus–Daphnia* interaction, including its basic parameterization. I then test its validity by comparing its predictions to the observed results of two laboratory experiments designed to investigate population growth of *D. pulex* under the influence of different levels of *Chaoborus* predation. Finally, I use the model to

simulate the effects of *Chaoborus* predation on different life history strategies of *Daphnia* in order to analyze the evolution of delayed reproduction in *D. pulex* as an induced response to this predator.

Daphnia and *Chaoborus*

Daphnia are small (1–3 mm adult body length) herbivorous suspension-feeders that are ecologically important in a variety of freshwater lentic habitats (Peters and de Bernardi, 1987). During 'favorable' periods of the year, these animals reproduce via parthenogenesis to produce subitaneous (immediately-hatching) eggs. The number of eggs in a clutch is determined by food abundance and the size of the adult female (Green, 1956; Hebert, 1978), while embryonic development time is largely dependent on water temperature (Vijverberg, 1980). Growth is indeterminate and limited to a short period immediately following a molt, resulting in a discrete pattern of size increase in which each instar maintains a constant body length until it molts to the next instar. Most *Daphnia* have 3–5 juvenile instars (Green, 1956) followed by several adult instars, each of which concludes with the release of a clutch of neonates. In *D. pulex*, neonates vary in body length from 0.5–0.8 mm, and mature in either the fifth or sixth (usually the fifth) instar (Green, 1956; Lynch, 1983, 1984; Dodson and Havel, 1988; Riessen and Sprules, 1990). Size at maturity varies from about 1.4–2.0 mm, while maximum body length can be 3.0 mm (Lynch, 1983, 1984; Riessen, 1994).

Chaoborus larvae are tactile, ambush predators (Greene, 1983) that are common in plankton communities of both lakes and ponds. They remain motionless in the water column, use mechanoreceptors to detect hydrodynamic disturbances made by approaching prey, and attack and capture these prey with their prehensile antennae and mandibles. There are four larval instars – the first two feed primarily on very small prey, such as rotifers and copepod nauplii, while instars 3 and 4 tend to select larger crustacean zooplankton as prey (Fedorenko, 1975; Moore, 1988; Swift, 1992). The fourth instar is the most voracious and longest lasting during the life cycle. *C. americanus* is one of the larger species, with fourth instars measuring about 10 mm in body length and having head lengths and mouth gapes of about 1.3–1.6 mm and 0.7 mm, respectively (Swift and Fedorenko, 1975; Fedorenko, 1975; personal observation).

Model development

General model

The basic model is a stage-classified matrix population model for *Daphnia pulex* (stages = instars), which uses the methods described by Caswell (1989)

along with the software program RAMAS/stage (Applied Biomathematics, Setauket, NY). The primary elements of the population projection matrix include fecundity at instar i (F_i) and the probabilities that *Daphnia* either survives and remains in instar i during a 1-d projection interval (P_i) or survives and grows to instar i + 1 during this interval (G_i). F_i is determined from the clutch sizes of the various adult instars ($F_i = 0$ for juvenile instars), while P_i and G_i are functions of both instar duration (T_i) and instar-specific survival probability (σ_i) (Caswell, 1989: Eqs. 4.60, 4.61, 4.77).

Modification of the instar-specific survival probability elements of the projection matrix account for the effects of *Chaoborus* predation. The survival probability for each instar during the 1-d projection interval (σ_i) is the product of a non-predation or 'background' survival probability (b_i) and the probability of survival in the face of *Chaoborus* predation (c_i). This latter probability is calculated following the procedures outlined by Riessen (1992), and is a function of (1) the encounter rate between *Daphnia* and *Chaoborus* (E_i), which determines the probability of an encounter between predator and prey, (2) the probability that an encounter results in a strike (attack) by the predator [*P*(Strike)], and (3) the probability that a strike results in successful ingestion of the prey (strike efficiency). Encounter rate is a function of *Chaoborus* density and *Daphnia* body length (Riessen, 1992: Eq. 2). *Daphnia* swimming speed increases with increasing body length, resulting in higher encounter rates with the stationary ambush predator (Pastorok, 1981; Riessen *et al.*, 1988). Strike efficiency is also a function of *Daphnia* body length, but with the opposite effect. Increasing *Daphnia* body size results in a lower strike efficiency by *Chaoborus*, since larger prey are more difficult to handle and ingest (Swift and Fedorenko, 1975; Pastorok, 1981; Riessen *et al.*, 1988; Swift, 1992; this study). The net result of these counteracting effects of *Daphnia* body length is *Chaoborus* selection for intermediate-sized *Daphnia* (Pastorok, 1981; Riessen, 1992).

Parameterization of the model requires estimates of body length, clutch size, development time (instar duration), and background (non-predation) mortality for each *Daphnia* instar, as well as *Chaoborus* density and values for *P*(Strike) and strike efficiency. Demographic predictions of the model include *Daphnia* population growth rate, λ (dominant eigenvalue) or r ($= \ln \lambda$), stable stage distribution (right eigenvector), and reproductive values (left eigenvector).

Parameter estimation for Chaoborus-*induced mortality*

Encounter rate (E_i) between the stationary fourth-instar *Chaoborus* and *Daphnia* of different body sizes was calculated from Riessen (1992: Eq. 2), which is a modified version of the relationship formulated by Pastorok (1981). Encounter probability was then calculated from the encounter rate using the Poisson distribution (see Riessen, 1992: Eq. 1).

Probability of a strike given an encounter [*P*(Strike)] and strike efficiency were determined from observations of predator-prey interactions between fourth-instar *C. americanus* (mean head capsule length = 1.31 mm) and nine size classes of *D. pulex*, ranging in body length from 0.70–1.85 mm (Fig. 1). Interactions in each of the nine trials occurred at room temperature (22–27 °C) in an 8-cm glass cube filled with 500 ml of aged tap water and containing a single *Chaoborus* and 15–30 *D. pulex*, depending on prey size. An encounter was judged to occur when a *Daphnia* approached within ≈3 mm of the body of the predator (Riessen *et al.*, 1984). *P*(Strike) is the proportion of encounters that resulted in strikes and strike efficiency is the proportion of strikes that resulted in successful ingestion by *Chaoborus*. Each trial with a given *Daphnia* body size used 5–10 *Chaoborus* that were starved for 2–3 days and produced from 79–148 total predator–prey encounters.

There was no effect of *Daphnia* body length on *P*(Strike) values within the size range of 0.70–1.67 mm (slope of linear regression not significantly different from 0) (Fig. 1A). At the largest *Daphnia* body length tested (1.85 mm), however, the probability of a strike given an encounter was reduced. After a few unsuccessful attempts by a *Chaoborus* to ingest these large prey, *Daphnia* that subsequently entered the predator's encounter volume were usually no longer attacked. This introduced a bias in the calculation of *P*(Strike) for 1.85-mm *Daphnia* (value influenced by previous unsuccessful experiences), and I therefore eliminated it from the analysis and used a value of *P*(Strike) = 0.727 (mean of values between 0.70–1.67 mm body length) for all *Daphnia* size classes. Since strike efficiency on the 1.85-mm *Daphnia* approaches 0 (see below), *Chaoborus*-induced mortality on these large prey is inconsequential and the exact value of *P*(Strike) for prey ≥1.85 mm becomes irrelevant.

Strike efficiency for fourth-instar *C. americanus* was inversely related to *Daphnia* body length (Fig. 1B). *D. pulex* up to and including 1.67 mm body length can be ingested, and the relationship between strike efficiency (S_i) and body length (L) is approximately linear:

$$S_i = -0.246\ L + 0.501. \quad (1)$$

At 1.85 mm body length, strike efficiency is near 0. No *Daphnia* of this size were actually ingested, but a single individual was killed after initial capture and later release, resulting in one death out of 74 strikes ($S_i = 0.014$). Since the upper limit for *Daphnia* body size that can be ingested by these *Chaoborus* lies between 1.67–1.85 mm, the strike efficiency function above 1.67 mm body length was estimated according to the following linear function (Fig. 1B):

$$S_i = -0.422\ L + 0.795. \quad (2)$$

Equation (1) is based on a linear regression ($n = 8$), while Equation (2) is based on an interpolation of strike efficiency between 1.67 mm [maximum body

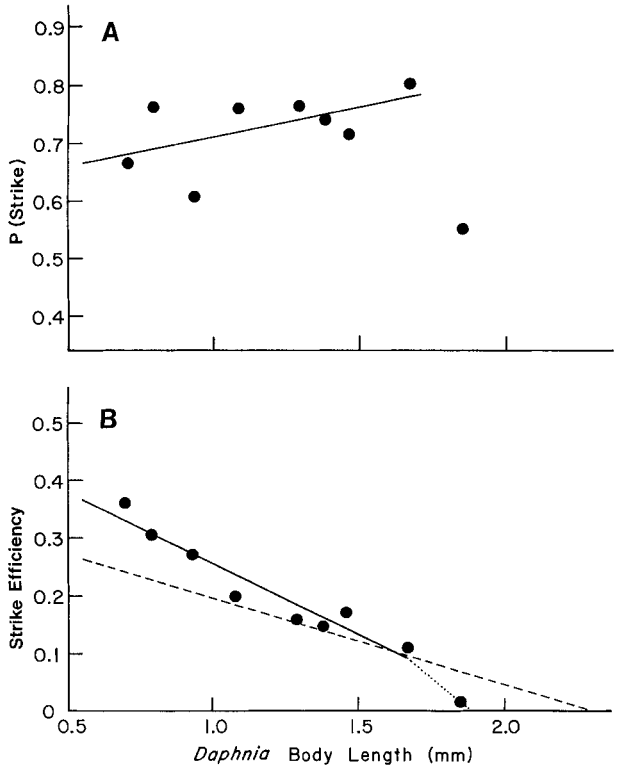


Figure 1. Predation components for interaction between fourth-instar *Chaoborus americanus* (mean head capsule length = 1.31 mm) and *Daphnia pulex* of various sizes. (**A**) Probability of a strike given an encounter [*P*(Strike)]. Solid line represents linear regression for *Daphnia* body lengths between 0.70–1.67 mm ($n = 8$): $y = 0.103x + 0.607$ ($r^2 = 0.313$, $p = 0.150$). Mean value of *P*(Strike) = 0.727 between 0.70–1.67 mm *Daphnia* body length. See text for further explanation. (**B**) Strike efficiency. Solid line represents linear regression for *Daphnia* body lengths between 0.70–1.67 mm ($n = 8$): $y = -0.246x + 0.501$ ($r^2 = 0.923$, $p = 0.0001$). Dotted line represents strike efficiency function above 1.67 mm body length: $y = -0.422x + 0.795$. Dashed line represents strike efficiency function from data of Swift and Fedorenko (1975) for larger fourth-instar *C. americanus* (mean head capsule length = 1.60 mm) preying on *Daphnia* between 0.6–2.2 mm body length ($n = 5$): $y = -0.151x + 0.346$ ($r^2 = 0.971$, $p = 0.0021$).

length in Equation (1)] and 1.85 mm (S_i near 0) (Fig. 1B). This two-part procedure produces more accurate estimates of strike efficiency for all body lengths than would be obtained by using a single linear regression that included the value of S_i at 1.85 mm body length. Including points where $S_i = 0$ in a single linear regression would overestimate the maximum size that can be ingested and distort the relationship for smaller body lengths that are vulnerable to ingestion. These functions [Equations (1) and (2)] are somewhat different than the one established by Swift and Fedorenko (1975) for larger fourth-instar *C. americanus* (mean head capsule length = 1.60 mm). These larger

C. americanus could consume larger *Daphnia* than the smaller *Chaoborus* I examined, but were less efficient in capturing and ingesting smaller *Daphnia* (Fig. 1B).

Kairomone-induced neck spines on juvenile *D. pulex* reduce the strike efficiency of *Chaoborus* to values between 25–50% of those on undefended individuals of the same body length (Krueger and Dodson, 1981; Havel and Dodson, 1984; Tollrian, 1995a). This is accounted for in the model [as a correction to Equation (1)] by reducing the *Chaoborus* strike efficiency calculated for juvenile *D. pulex* in Equation (1) by the appropriate amount when these individuals have neck spines. This reduced strike efficiency alters values of c_i and thus survival probability for that instar in the projection matrix.

Model validation

I tested this demographic model with two experiments (experiments 1 and 2) that examined the effect, over a 7-d period, of *Chaoborus* density on the population growth of *D. pulex* in 4.0-l beakers. Five different predator treatments (four replicates each) were established by adding fourth-instar *C. americanus* (same size as those used previously; mean head capsule length = 1.32 mm) to the beakers in densities representing the range found in nature: 0/beaker (control), 1/beaker (0.25/l), 2/beaker (0.5/l), 4/beaker (1.0/l), and 8/beaker (2.0/l). At the end of 7 days the contents of each beaker were filtered through 110-μm Nitex netting and preserved in 6–8% Formalin solution containing sucrose (Haney and Hall, 1973). I tested the model by comparing its predictions for *D. pulex* population growth (final densities on Day 7) with the results observed in each predator treatment of both experiments (Figs. 2–3). Knowledge of the abundances of the various *Daphnia* instars at the start of an experiment (see Appendix), along with estimates for F_i (clutch sizes), T_i (instar durations), b_i (background survivorship), and instar-specific body lengths and neck spine development (used in the calculation of c_i), allow the model to predict population density at any point in time thereafter, assuming that environmental conditions (e.g., water temperature, food, *Chaoborus* density) have remained constant. Details of the methodology involved in each experiment, as well as that used to estimate model parameters for this particular test, are provided in the Appendix. All statistical analyses were performed using the software program Statistix for Windows (Analytical Software, Tallahassee, FL).

In both experiments, increasing levels of *Chaoborus* predation resulted in a progressive and significant decrease in numbers of *Daphnia* in the experimental beakers (One-way ANOVA on the five predation treatments: $p < 0.0001$ for experiment 1, $p = 0.0003$ for experiment 2) (Fig. 2). This decrease in *Daphnia*

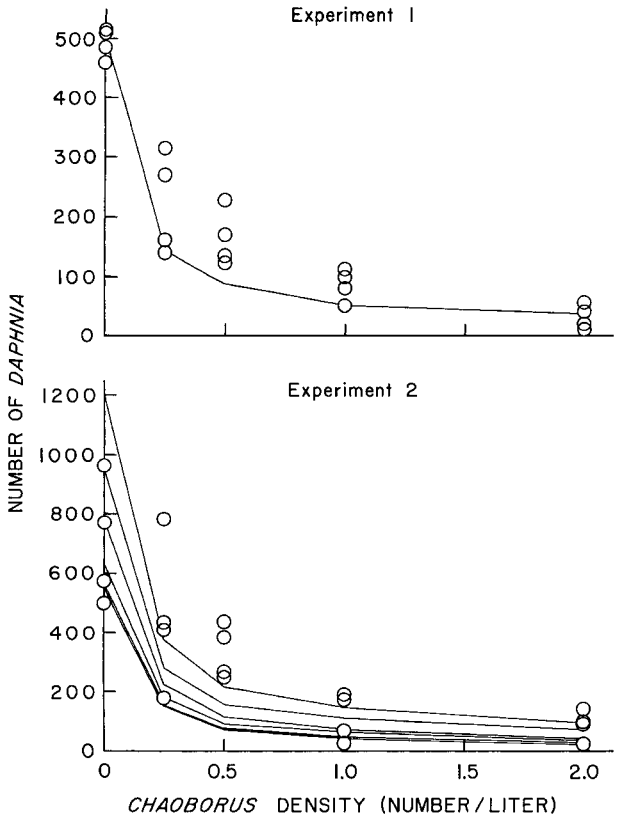


Figure 2. Results of experiments 1 and 2 (total number of *Daphnia pulex* per 4-l beaker at end of 7-d experiment) compared to model predictions for each predation treatment (*Chaoborus*/l). (○) Observed results for each treatment in each experiment. (——) Model predictions. The six lines in experiment 2 represent trajectories resulting from different initial *Daphnia* population conditions, and thus display the range of model predictions for this experiment. See text for statistical analysis.

population size was accurately predicted by the model in all but one treatment in each experiment (Fig. 2). In experiment 1, the single predicted value of the model for each predation treatment (*Chaoborus* density) was compared to the four observed values for that treatment using a one-sample *t*-test. There were no significant differences between observed and predicted values ($p > 0.05$) in the various predation treatments, except for the one represented by 0.5 *Chaoborus*/l ($p = 0.0476$). In experiment 2, the six predicted model values (representing the predicted outcomes from the range of initial *Daphnia* population conditions; see Appendix) for each predation treatment were compared to the four observed values using a two-sample *t*-test. Again, there were no significant differences between observed and predicted values ($p > 0.05$), except for the 0.5 *Chaoborus*/l treatment ($p = 0.0120$). Thus, in each experiment, the

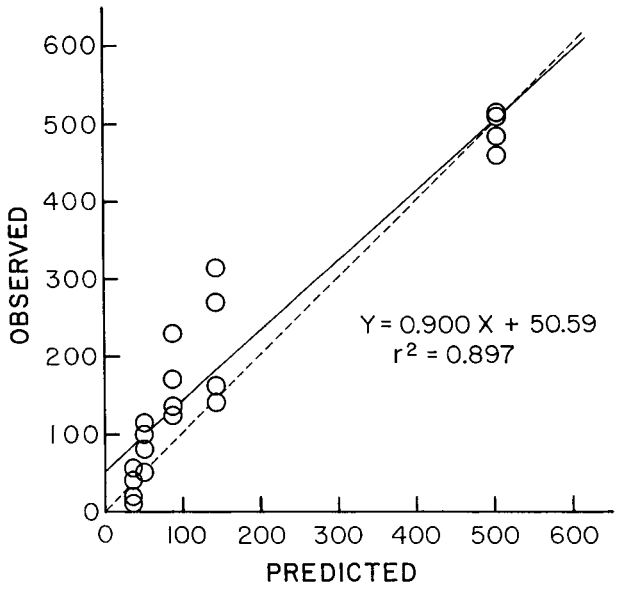


Figure 3. Linear regression (solid line) of observed number of *Daphnia* on number predicted by model: $y = 0.900x + 50.59$ ($r^2 = 0.897$). Dashed line represents results of a 'perfect' model, with slope = 1.0 and intercept = 0. See text for statistical analysis.

observed results (total number of *Daphnia*) were not significantly different from the model predictions in four of the five predation treatments.

The results of experiment 1 were further analyzed with a linear regression of the observed values on the model predictions (Fig. 3). A 'perfect' model would produce a regression line with a slope of 1.0 and an intercept of 0 (Haefner, 1996). The slope of the regression line in this analysis was 0.900 (95% Confidence Interval (CI) = 0.749–1.052), which was not significantly different from a slope of 1.0 ($p > 0.05$), but was significantly different from a slope of 0 ($p < 0.0001$). The intercept of the regression line (50.59), however, was significantly different from 0 ($p = 0.0090$). While the intercept deviated from 0 (mainly due to results from the 0.5 *Chaoborus*/l treatment), the close approximation of the slope to 1.0 indicates that the reduction in number of *Daphnia* with increasing levels of *Chaoborus* predation pressure (a critical aspect of the model) is closely approximated by the model predictions.

The model somewhat overestimates the impact of *Chaoborus* predation (i.e., underestimates the number of *Daphnia*) in the 0.5 *Chaoborus*/l treatment and, to a lesser extent, the 0.25 *Chaoborus*/l treatment in both experiments (Fig. 2). This discrepancy between observed and predicted values in the low predation treatments may be due to possible satiation of *Chaoborus* during the course of the experiment, which would result in reduced feeding and thus a lower predation impact than would otherwise occur. This disparity only occurs in the

low *Chaoborus* density treatments, where *Daphnia* densities are relatively high, making predator-prey encounters more numerous and satiation of individual *Chaoborus* more likely. Thus, while the model slightly underestimates *Daphnia* numbers in the low *Chaoborus* treatments, this may be an artifact of the experimental design of the test and not a problem with the model itself.

In any event, despite this slight discrepancy, these experiments provide reasonable validation for the model and its ability to predict the impact of *Chaoborus* predation on *Daphnia* population dynamics.

Evolution of delayed reproduction in *Daphnia*

I use the model developed above to examine the evolution of delayed reproduction in *D. pulex* as an induced response to *Chaoborus* predation. I specifically investigate the feasibility of the hypothesis proposed by Black (1993) and Tollrian (1995b) that this response is linked with increased body size as part of an adaptive life history shift that involves a trade-off between the two traits. In particular, this hypothesis proposes both that (1) an increase in *Daphnia* body length will result in reduced vulnerability to *Chaoborus* predation, and (2) such a reduction in predator-induced mortality will compensate for the demographic disadvantage of delayed reproduction. Fitness values (r) are calculated in simulations that represent life histories exhibiting different developmental rates ('normal' or 'delayed' reproduction) and varying degrees of increase in body length (measured as a fixed percent increase for each instar in the growth trajectory) for *D. pulex* subjected to moderate levels of *Chaoborus* predation pressure. I make these comparisons separately for *D. pulex* of three different growth trajectories (body size patterns), representing the range of body size exhibited by this species in nature, and for two different *Chaoborus* strike efficiency patterns, representing 'smaller' and 'larger' fourth-instar *C. americanus*.

Model parameterization

This version of the model is comprised of 15 instars (stages) for *D. pulex*, with maturity being reached in the fifth instar. In all simulations, the *Chaoborus* density was set at 0.5/l, a value which is commonly observed in nature and which represents a moderate level of predation pressure. Encounter probability between various *Daphnia* instars and *Chaoborus* was calculated as described previously (see Riessen, 1992: Eqs. 1, 2), and the probability of a strike given an encounter [P(Strike)] was assumed to be 0.727 for all *Daphnia* size classes, as determined above (Fig. 1A).

Simulations were run for three base *Daphnia* growth trajectories, with primiparous and maximum body lengths of 1.40/2.00 mm (small),

Table 1. Body lengths in the three simulated growth trajectories (Small, Medium, and Large), fecundity (F_i), instar durations (T_i, in days), and background survivorship values (b_i) for the 15 *Daphnia pulex* instars (first adult instar = instar 5) present in the matrix population model used to examine life history strategies of *D. pulex* in response to *Chaoborus* predation

Instar	Body length (mm)			F_i	T_i	b_i
	Small	Medium	Large			
1	0.63	0.65	0.67	–	1.20	1.000
2	0.76	0.83	0.90	–	1.20	1.000
3	0.95	1.09	1.23	–	1.20	1.000
4	1.18	1.41	1.63	–	1.20	1.000
5	1.40	1.70	2.00	5	2.40	0.999
6	1.52	1.84	2.16	9	2.45	0.997
7	1.60	1.94	2.27	12	2.55	0.995
8	1.66	2.01	2.35	14	2.55	0.990
9	1.72	2.08	2.43	15	2.55	0.980
10	1.78	2.15	2.51	15	2.55	0.970
11	1.84	2.21	2.58	15	2.55	0.965
12	1.89	2.27	2.65	15	2.55	0.960
13	1.93	2.32	2.70	15	2.55	0.955
14	1.97	2.36	2.75	15	2.55	0.950
15	2.00	2.40	2.80	15	2.55	0.950

1.70/2.40 mm (medium), and 2.00/2.80 mm (large) (Table 1). This represents the range of body sizes exhibited by *D. pulex* (Lynch, 1989; Riessen, 1994; this study), and allows an analysis of the effect of *D. pulex* body size on the feasibility of a life history trade-off between further increases in body size and delayed reproduction.

Fecundity (F_i) of *D. pulex* was assumed to be the same for all growth trajectories (i.e., constant for a given instar, regardless of its body length), and increased from 5 eggs/clutch in the primiparous (fifth) instar up to a maximum of 15 eggs/clutch (Table 1). These represent moderate values for *D. pulex* clutch sizes, well below those attainable under optimal food conditions (see Lynch, 1989; Riessen and Sprules, 1990; this study). Although the clutch size of a given adult instar does tend to increase with increasing body length of that instar, I make the assumption of constant fecundity because these simulations are designed solely to examine the direct effect of body size increase on the vulnerability of *Daphnia* to *Chaoborus*, and to specifically test the hypothesis of Black (1993) and Tollrian (1995b) on the evolution of delayed reproduction in *D. pulex* as an induced response to *Chaoborus* predation.

Durations of the various *D. pulex* instars (T_i) were also assumed to be the same for all three growth trajectories (Table 1), and represent values typical at $\approx$20 °C in the absence of *Chaoborus* kairomones (Lynch, 1989; this study). In those simulations in which developmental delay is exhibited due to the presence of *Chaoborus* kairomones (Fig. 4: closed circles), these instar durations were

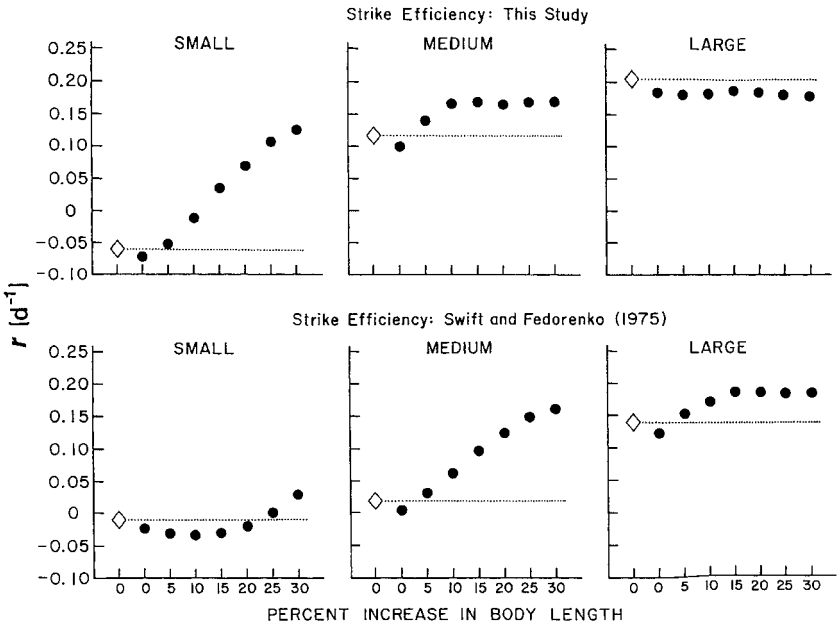


Figure 4. Model predictions of changes in *Daphnia pulex* fitness (measured as population growth rate, *r*) with increasing body length under a simulated predation pressure of 0.5 *Chaoborus*/l. Increasing body length is measured as percent increase in each instar from three base growth trajectories (Table 1): small (left panels), medium (middle panels), and large (right panels). Top panels simulate strike efficiency pattern of 'smaller' *Chaoborus americanus* described in this study (mean head capsule length = 1.31 mm), while bottom panels simulate strike efficiency pattern of 'larger' *C. americanus* described by Swift and Fedorenko (1975) (mean head capsule length = 1.60 mm) (Fig. 1B). (◇, ...) Fitness values at base growth trajectories (0% increase) with no delayed development (see text). (●) Fitness values with delayed development characteristic of response of *D. pulex* to *Chaoborus* kairomones (see text). The difference between values at the open diamond/dotted line and first closed circle (0% increase in body length) in each simulation is the cost of delayed development. Subsequent closed circles represent increased body length (fixed percentage for each instar). In those that exceed the level represented by the dotted line, the benefit of reduced mortality from *Chaoborus* predation (due to increased body length) exceeds the cost incurred by delayed development. In those that remain below the dotted line, the cost exceeds any benefit.

increased by 12% for juveniles (instars 1–4) and 3% for adults (instars 5–15) (Riessen and Sprules, 1990; Tollrian, 1995b).

Background (non-predation) survivorship (b_i) was estimated from the survivorship component of the life table experiments of Riessen and Sprules (1990: Fig. 2B), which examined *D. pulex* at 22 °C under near optimal food conditions (1×10^5 cells/ml *Chlamydomonas*). Based on these data, survivorship (probability of surviving during a 1-d time interval) for all three growth trajectories was assumed to be 1.00 for all four juvenile instars and to progressively decrease to 0.95 by instar 15 (Table 1). These values are also consistent with the pattern of *D. pulex* survivorship reported by Paloheimo and Taylor (1987).

Strike efficiency for *Chaoborus* on *Daphnia* of various body sizes was calculated using both the pattern generated in this study [Equations (1), (2)] for 'smaller' fourth-instar *C. americanus* (mean head capsule length = 1.31 mm) and that determined from the data of Swift and Fedorenko (1975) for 'larger' individuals of this species from another population (mean head capsule length = 1.60 mm) (Fig. 1B). Strike efficiency values on small *Daphnia* are lower for the 'larger' *C. americanus* than for the 'smaller' predators, but the maximum prey size that can be consumed is greater for the 'larger' predators (Fig. 1B). Simulations using both of these strike efficiency patterns, as well as the three base *Daphnia* growth trajectories, were performed to illustrate the effects of relative sizes of predator and prey on the pattern of *Daphnia* life history evolution. In all of these simulations I assumed that *D. pulex* instars 2–4 have neck spines [neonates often have very reduced neck spine development (Tollrian, 1993, 1995a; this study)], and that these defenses reduced *Chaoborus* strike efficiency values by 50% (see Model development). An additional series of simulations in which neck spines were assumed to be present only on *D. pulex* instars 2–3, which usually exhibit maximum neck spine development (Havel and Dodson, 1984; Tollrian 1993, 1995a; this study), produced results that are qualitatively the same as those described below.

Simulation results

The results of the life history simulations, and thus the feasibility of selection for increased body size and delayed reproduction, are greatly dependent on the relative sizes of predator and prey. In four of the six *Daphnia–Chaoborus* interactions modeled (Fig. 4), an increase in *Daphnia* body length of only 5% provides sufficient benefit in reduced vulnerability to *Chaoborus* to compensate for the cost of delayed reproduction, thus providing an adaptive advantage for this life history trade-off. This is the case for small to medium-sized *D. pulex* interacting with relatively small *C. americanus* and for medium to large-sized *D. pulex* under predation by larger *C. americanus*. In these situations at least some of the adult *Daphnia* (especially instars 5–6, the smallest adults) are in size classes that make them highly vulnerable to *Chaoborus* predation. A general increase in body length for all instars results in an increase in survivorship for adults, which is substantial for some instars, as they become more difficult for *Chaoborus* to ingest, but a modest decrease in survivorship for most juveniles as they grow to more vulnerable sizes. The net effect, however, is an increase in fitness, which even at a 5% increase in body length exceeds the cost of delayed reproduction. The 6–9% increase in size at maturity observed by Black (1993) and Tollrian (1995b) for *D. pulex* exposed to *Chaoborus* kairomones would therefore be sufficient to favor this trade-off under these circumstances.

In the other two simulation conditions, however, increased body length provides no benefit to *Daphnia*, which would select against the evolution of delayed reproduction as a trade-off for increased body size (Fig. 4). When large *D. pulex* are subject to predation from smaller *C. americanus* (Fig. 4: top right panel), all of the adult instars are so large, relative to predator size, that they are invulnerable to *Chaoborus* and further increases in body length provide no additional benefit. When small *D. pulex* interact with the larger *C. americanus* (Fig. 4: bottom left panel), an increase in body length at first reduces fitness before it eventually increases to values greater than that of the base growth trajectory with no delayed development. All adult instars are highly vulnerable to predation and increases in body length decrease this vulnerability and increase survivorship. At small to moderate increases in body length ($\leq$15%), however, this effect is exceeded by the increased vulnerability that occurs in juveniles as they become larger, resulting in decreased fitness. This outcome is reversed when body length increases are larger. The evolution of a life history pattern that trades delayed reproduction for increased body size would be difficult in this situation since the benefits would not exceed the costs until the *Daphnia* had increased in body length by at least 25%. Initial increases in body length from the base growth trajectory would create an 'adaptive valley' that would be difficult to cross before attaining the benefits of very large body size increases.

Discussion

Models previously developed to investigate the influence of size-selective predation on life history evolution in *Daphnia* and other zooplankton (Lynch, 1980; Taylor, 1980; Taylor and Gabriel, 1992) have been general in nature, and limited to the comparison of idealized patterns of predation by fish (select large prey) and carnivorous invertebrates (select small prey). These models do not examine the patterns and effects of specific predators, and thus are limited in their usefulness for understanding the impacts of any particular predator on its prey. The model I develop in this paper, on the other hand, is specific for the interaction between *Chaoborus americanus* and *Daphnia pulex*, although it can easily be modified to analyze any *Chaoborus–Daphnia* interaction. The generality that is lost in this type of model is compensated by a gain in realism and precision (Levins, 1966) that allows the model to accurately determine the impact of *Chaoborus* predation on the population dynamics and life history of *Daphnia*.

The precise predictions made by the model also allow it to be accurately tested. In this study, both experiments designed as a test for the model (experiments 1 and 2) provided strong validation for its ability to accurately

predict the degree to which *Chaoborus* predation affects *Daphnia* population growth. This in turn provides validity for the model's application in investigating the impact of *Chaoborus* predation on *Daphnia* life history evolution. In addition to estimating the short-term population dynamics of *Daphnia* exposed to different levels of *Chaoborus* predation and the fitness of various life histories, the model can be used to analyze the costs and benefits of morphological (e.g., neck spine formation) and behavioral (e.g., vertical migration) changes in *Daphnia* (see Riessen, 1992). It could also be expanded to predict long-term *Daphnia* population dynamics by allowing for seasonal changes in clutch size and growth rate (influenced by changing food levels), development rate (influenced by changing water temperature), and *Chaoborus* density.

In this study, I apply the model to investigate the evolution of delayed reproduction in *Daphnia* that are exposed to *Chaoborus* predation. The outcome of these simulations supports the hypothesis (Black, 1993; Tollrian, 1995b) that under certain circumstances delayed reproduction in *Daphnia*, which by itself results in a decrease in fitness, could evolve in response to *Chaoborus* predation as part of an integrated suite of life history traits that trades off delayed reproduction for increased body size. This outcome demonstrates that the benefit to *Daphnia* of increased body size (i.e., reduced mortality from *Chaoborus* predation), even at increases of only 5% from the base growth trajectory, often exceeds the cost of delayed reproduction.

Delaying maturity generally reduces reproductive output and values of r, and thus has the effect of decreasing fitness, however there are circumstances under which this type of change may be favored by natural selection. These typically involve trade-offs that result in either higher initial fecundity or higher juvenile survivorship to a degree that compensates for the demographic cost of delayed reproduction (Stearns, 1992). The results of this present study indicate that delayed reproduction in *Daphnia* may evolve as part of an integrated life history that involves increasing body size as a means to reduce *Chaoborus* predation. Similar life history responses and trade-offs (larger body size to reduce predation at the expense of delayed reproduction) have been documented by Crowl and Covich (1990) for snails exposed to size-selective crayfish predation, and Belk (1998) for some populations of bluegill sunfish living in the presence of predatory largemouth bass. In addition, Law (1979) and Michod (1979) have developed life history models that examine the general effects of age-specific mortality on reproductive effort and the evolution of delayed reproduction. Other general life history models have investigated additional theoretical conditions under which delayed reproduction may evolve (Stearns and Crandall, 1981; Tuljapurkar, 1990; Stearns, 1992; Takada and Caswell, 1997; Van Dooren and Metz, 1998).

Some outcomes of the model (2 of the 6 conditions), however, do not favor the evolution of the proposed trade-off of delayed reproduction for increased

body size in *D. pulex* exposed to *Chaoborus* predation. Indeed, if this trade-off was a general life history phenomenon, there should be a tight coupling of these two life history traits. A review of 14 studies that examined the effects of *Chaoborus* kairomones on *D. pulex* life history parameters (Tollrian, 1995b: Table 8) indicates a lack of such a coupling. Only four of those studies observed an increase in *Daphnia* body length at maturity, and only two of these (Black, 1993; Tollrian, 1995b) were associated with delayed reproduction. While delayed reproduction was a common response to *Chaoborus* kairomones (10 of the 14 studies), it was usually not associated with increased body size (2 of the 10 cases). In addition, Spitze (1991) observed that *Chaoborus* predation on laboratory populations of *D. pulex* resulted in the evolution of both larger *Daphnia* body size and earlier reproduction. Delayed reproduction, therefore, does not always accompany the evolution of increased body size.

Clearly, the evolution of particular life history responses by *D. pulex* to *Chaoborus* predation involves a complex set of interactions which are determined by predator and prey body sizes (this study), environmental variables (e.g., food levels, water temperature), and other defensive responses (morphological and behavioral) of *D. pulex*. These interactions may result in several different adaptive suites of traits exhibited by different *D. pulex* clones under various sets of circumstances (see Tollrian, 1995b: Table 8). Deciphering the complexity of these ecological interactions and evolutionary responses will require ever more sophisticated models that include not only the precise relationship between *Daphnia* body size and vulnerability to *Chaoborus*, but also the influence of nutrition, water temperature, and other predators. The approach in model development provided in this study is a step in that direction.

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Appendix: test of the model

Experimental methodology

Each experiment was initiated by adding 23–48 *D. pulex* to each of twenty 4.0-l beakers (5 predator treatments × 4 replicates each) containing aged tap water,

1×10^5 cells/ml *Scenedesmus* for food, to allow for maximal population growth of *Daphnia* (Porter *et al*. 1983), and 0–8 fourth-instar *Chaoborus americanus* larvae. All *D*. *pulex* were from the same clone, which was originally isolated from a temporary pond in Erie County, New York. *C*. *americanus* larvae were collected from a permanent pond in Niagara County, New York one week before the beginning of each experiment, and were starved for either one (experiment 1) or two (experiment 2) days before the experiment. All beakers were incubated in an environmental chamber (Percival Model I-37LL) for 7 days at 20 °C with a 14:10 L:D photoperiod. The two experiments differed in initial densities and instar distributions of the *Daphnia*. In experiment 1, the *D*. *pulex* populations were each started with 23 individuals collected from a stock population that was cultured under food, temperature, and light conditions similar to that of the experiment. *Daphnia* in this stock population were well fed, and adults carried several eggs in their brood chambers. The 23 initial *Daphnia* included 8 small juveniles (instars 1 and 2), 7 large juveniles (instar 4), 2 small adults (instar 5), and 6 large adults (instars 7–9) (Table A.1). This created similar starting conditions for each of the twenty populations, although clutch sizes undoubtedly varied somewhat for the adults. An extra set of 23 *Daphnia* was preserved to precisely determine the initial instar distribution of the starting populations.

In experiment 2, the initial *Daphnia* population was selected to more closely mimic that of an exponentially growing population. This was accomplished by

Table A.1. Initial instar distributions in experiments 1 and 2. Each was used to initiate a set of model simulations for that experiment. The six replicate distributions in experiment 2 estimate the variability present among beakers in starting population conditions. Values are numbers of *Daphnia* (percent) that each instar contributes to the starting population in each experiment, including the six replicates in experiment 2

Instar	Experiment 1	Experiment 2						
		1	2	3	4	5	6	(Mean %)
1	6 (26.1)	4	2	7	13	7	8	(18.3)
2	2 (8.7)	14	14	17	10	11	13	(36.8)
3	0	5	6	6	4	4	8	(15.1)
4	7 (30.4)	3	3	6	3	6	5	(11.7)
5	2 (8.7)	1	2	0	1	2	5	(4.9)
6	0	2	2	5	2	2	7	(8.5)
7	2 (8.7)	0	0	0	1	0	1	(0.8)
8	1 (4.3)	1	0	0	0	0	0	(0.6)
9	3 (13.0)	0	0	2	0	2	0	(1.7)
10	0	0	0	0	1	1	0	(1.0)
11	0	0	0	1	0	0	0	(0.4)
12	0	0	0	0	0	0	1	(0.4)
13	0	0	0	0	0	0	0	(0)
Total	23 (100)	30	29	44	35	35	48	(100)

adding one 521-ml Fleaker sample of a rapidly growing stock population of *D. pulex* (from the middle of the aquarium after thorough mixing) to each beaker. The stock population, started 7 days before the experiment with *D. pulex* that had previously been cultured under high algal densities, was maintained at 20 °C and 14:10 L:D photoperiod with *Scenedesmus* densities that fluctuated between 5×10^4 and 2×10^5 cells/ml. Initial population sizes ranged from 29–48 *Daphnia*/beaker, with the great majority of individuals being juveniles (instars 1–4) as would be expected in an exponentially growing population (Table A.1). Estimates of starting population densities and instar distributions were obtained from six extra samples from the stock population that were preserved before the experiment. Thus there was greater variability among beakers in initial densities and instar distributions in experiment 2 than in experiment 1. This subsequently resulted in greater variability in final *D. pulex* densities within predator treatments in experiment 2.

Scenedesmus density and the condition of the *Chaoborus* were monitored daily during the course of the experiments. *Scenedesmus* densities increased in all beakers during both experiments. No differences among predator treatments were observed early in the experiments, but by day 6 in experiment 1 and day 4 in experiment 2 *Scenedesmus* density was higher in the 4 and 8 *Chaoborus*/beaker treatments than in the control. This was probably due to lower *Daphnia* densities in the high predator treatments, which resulted in reduced levels of herbivory on the algal population. By the end of experiment 1, mean *Scenedesmus* densities were 2.1×10^5 cells/ml in control beakers and 4.3×10^5 cells/ml in beakers with 8 *Chaoborus*. In experiment 2 on day 7, mean algal densities were 2.1×10^5 cells/ml in control beakers and 3.7×10^5 cells/ml in the beakers with highest predator densities. While food availability for *Daphnia* in the presence of high *Chaoborus* densities was higher in the latter part of each experiment, food levels in general were very high and probably beyond the point where such differences would appreciably affect ingestion rates. In any event, these differences in algal densities did not affect either clutch sizes or instar body lengths of *Daphnia* in different predator treatments.

Most of the *Chaoborus* present at the beginning of each experiment lived to the end and many had food in their crops each day. A few *Chaoborus* either died or appeared unhealthy, and these were replaced each day with new fourth-instar larvae. Of the total of 60 *Chaoborus* used in each experiment, an average of only 1.3 (experiment 1) and 0.7 (experiment 2) needed to be replaced each day.

At the end of each experiment, *Daphnia* population size and size distribution were determined in each beaker by examining the preserved animals at 25× or 50× under a stereomicroscope with an ocular micrometer. The presence or absence of neck spines on the juvenile *D. pulex* was also determined for each of the four juvenile instars.

Parameter estimation

I estimated body length and degree of neck spine induction (used in the calculation of c_i), clutch size (F_i), instar duration (T_i), and background (non-predation) survivorship (b_i) for each *Daphnia* instar in the *D. pulex* populations of both experiments. Mean body length for each instar (Table A.2) was determined from size–frequency distributions of the preserved *Daphnia* in each experiment. There were no differences in instar body length among treatments within an experiment, but *Daphnia* of a given instar (except instar 1) were consistently larger in experiment 2 (Table A.2). The means and standard deviations for clutch sizes of adult instars in each experiment (Table A.2) were estimated by determining the number of eggs or embryos in the brood chambers of preserved adult *Daphnia*, and assigning the clutch size of an individual to a particular adult instar based on the individual's body length. Clutch sizes were generally greater for a given adult instar in experiment 2 (Table A.2), which is consistent with the higher growth rates observed in this experiment.

The percentage of individuals in each juvenile instar (1–4) with well-developed neck spines was determined for each treatment (Table A.3) and used to

Table A.2. Body lengths, fecundity (F_i), instar durations (T_i), and background survivorship values (b_i) for the 13 *Daphnia pulex* instars present in the matrix population model for simulations of experiments 1 and 2

Instar	Body length (mm)		F_i[a]		T_i[b]	b_i
	Experiment 1	Experiment 2	Experiment 1	Experiment 2		
1	0.60	0.61	–	–	1.217	1.00
2	0.76	0.81	–	–	1.217	1.00
3	0.99	1.10	–	–	1.217	1.00
4	1.25	1.44	–	–	1.217	1.00
5	1.56	1.77	6.5 (1.2)	8.8 (2.3)	2.452	0.98
6	1.78	1.92	9.6 (0.8)	9.9 (3.0)	2.476	0.96
7	1.93	2.11	14.7 (1.6)	15.1 (5.2)	2.552	0.92
8	2.08	2.21	16.9 (2.5)	20.1 (5.0)	2.552	0.85
9	2.18	2.37	20.0 (4.1)	27.6 (6.2)	2.552	0.75
10	2.28	2.49	20.0 (5.1)	32.4 (7.1)	2.552	0.75
11	2.35	2.62	19.6 (5.0)	42.0 (6.3)	2.552	0.75
12	2.43	2.73	25.0 (3.0)	43.0 (6.5)[c]	2.552	0.75
13	2.52	2.85	27.9 (0)	43.0 (6.5)[c]	2.552	0.75

[a] Mean fecundity values (standard deviations shown in parentheses) were determined from clutch sizes of preserved *Daphnia* in experiments 1 and 2, reduced by 3.8% (value determined from experiment 4) to account for mortality of developing embryos (stillborn neonates).

[b] Instar durations (days) for juvenile instars (1–4) are mean values determined by dividing age at maturity by 4, while those for instars 7–13 are based on the mean value determined for instars 7 and 8.

[c] Clutch sizes for instars 12 and 13 in experiment 2 averaged together due to small sample size ($n = 3$ for both combined).

calculate the degree of reduction in *Chaoborus* strike efficiency due to this antipredator defense. Neck spines were not present on *D. pulex* in the controls (0 *Chaoborus*/l), except for a very few neonates in experiment 2. Development of this antipredator defense tended to be greatest in instars 2 and 3, and to increase with *Chaoborus* density (and thus kairomone concentration). It also was greater, for the most part, in experiment 1 than in experiment 2. I assumed that *Chaoborus* strike efficiency on spined individuals was reduced to 37.5% of that for undefended individuals (mean value between 25–50%; see Krueger and Dodson, 1981; Havel and Dodson, 1984; Tollrian, 1995a) and modified this value by the percentage of spined individuals present in an instar for a given treatment (Table A.3). Thus, for example, 63.2% of instar 3 individuals in the 0.25/l treatment of experiment 1 were spined, which reduced average *Chaoborus* strike efficiency on *D. pulex* of this size (0.99 mm body length) from 0.258 to 0.156 (strike efficiency would have been reduced to 0.097 if all individuals were spined). This degree of reduction translated into an increase in the 1-d survival probability (c_i) for this *D. pulex* instar from 0.777 to 0.858.

Instar durations (T_i) and background (non-predation) survivorship (b_i) for the various *D. pulex* instars were determined in two other experiments (experiments 3 and 4), which used the same *D. pulex* clone and the same environmental conditions as in experiments 1 and 2 (20 °C, 1×10^5 cells/ml *Scenedesmus* as food, 14:10 L:D cycle). In experiment 3, a cohort of five neonates was followed to maturity in the fifth instar and one adult raised for two additional instars, while in experiment 4, two cohorts of ten neonates each were followed through instars 6 and 9, respectively. Mean duration of juvenile instars (1–4) was determined by dividing the age at maturity (time from birth to when the first eggs are laid in the brood chamber at the beginning of the fifth instar) by 4 (number of juvenile instars). Mean duration of adult instars was estimated by determining the time between the release of successive clutches. Estimated values of T_i (Table A.2) were very similar to those reported by Lynch (1989) for *D. pulex* raised under similar environmental conditions. In all model simulations for treatments with *Chaoborus* present, these instar duration values were increased by 12% for juvenile instars and 3% for adult instars to account for the developmental delay that occurs in response to the presence of this predator (Riessen and Sprules, 1990; Tollrian, 1995b). Values of b_i used in the model simulations (Table A.2) reflect a pattern (100% juvenile survivorship followed by progressively decreasing adult survivorship) that closely approximates survivorship in experiments 3 and 4, although b_i values for instars 5–7 were slightly decreased to reflect the assumption of a smooth reduction in adult survivorship and to precisely calibrate the model under control (no *Chaoborus* predation) conditions.

Model simulations of the *D. pulex* populations in experiments 1 and 2 were initiated with the starting instar distributions present in each experiment

Table A.3. Percentage of *Daphnia pulex* individuals in juvenile instars 1–4 that have well-developed neck spines in the various treatments of experiments 1 and 2. Sample size given in ( )

Treatment (*Chaoborus*/l)	*Daphnia pulex* instar							
	Experiment 1				Experiment 2			
	1	2	3	4	1	2	3	4
0	0 (96)	0 (95)	0 (64)	0 (86)	2.8 (176)	0 (67)	0 (50)	0 (46)
0.25	41.4 (58)	49.0 (51)	63.2 (19)	31.6 (57)	33.9 (109)	0 (22)	20.5 (39)	0 (10)
0.5	35.2 (54)	92.1 (76)	90.9 (44)	62.5 (24)	27.1 (107)	32.3 (31)	58.1 (43)	0 (9)
1.0	76.0 (50)	98.0 (50)	83.3 (18)	82.5 (40)	20.5 (127)	77.8 (27)	86.7 (15)	75.0 (4)
2.0	40.0 (65)	100 (12)	100 (3)	37.5 (8)	34.1 (258)	83.3 (24)	100 (27)	[a]

[a] No fourth-instar *D. pulex* present in highest predation treatment of experiment 2. Neck spine induction assumed to be 75.0%, the same as in the next lowest predation treatment.

(Table A.1), as well as the various model parameters (F_i, T_i, b_i, c_i), and run for 7 time steps (each representing a 1-d interval) to mimic the duration of each experiment. In experiment 1, a single initial instar distribution represents the nearly uniform starting conditions present in this experiment (Table A.1). In experiment 2, each set of simulations was run six times, each representing the instar densities present in one of the six extra samples preserved to estimate initial population conditions (Table A.1). These six simulations were used to encompass the variability present among beakers at the start of the experiment.

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